



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations
 Summer Semester September 2023
22MCA101 - DATA STRUCTURES WITH ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Unit - I			
		Marks	BT*	CO*	PO*
1.	a) Define Abstract Data Type. Implement an ADT for rational numbers.	10	L2	1	2
	b) Define the term Array. How are two-dimensional arrays represented in memory? Explain how address of an element is calculated in a two dimensional array.	10	L2	1	1
2.	a) Write an algorithm to evaluate an expression given in postfix notation. Trace your algorithm for the following expression AB^CD-EF/HG+/*	10	L2	1	2
	b) Consider an arithmetic expression which includes {, [and (with the corresponding closing pairs. Write a program to check if the given expression contains a matching pairs of brackets. Trace with an example.	10	L2	1	1
		UNIT-II			
3.	a) Write a C program to simulate Circular queue. Trace the program for queue size=8.	10	L2	2	2
	b) List the difference between ordinary queue and circular queue. Write a C program to simulate Ordinary queue.	10	L2	2	2
4.	a) Describe the efficiency of recursion. Write a recursive C Program to find n th Fibonacci number and trace it for n=3	10	L2	2	2
	b) Define Recursion. Write the differences between iterative approach and recursive approach with an example.	10	L2	2	2
		UNIT-III			
5.	a) Write a C code for the following: i) Delete a node (whose info field is specified) from an ordered linked list ii) Insert a node into an ordered linked list.	10	L2	3	2
	b) Define Circular linked list. Implement the Queue operations using Circular linked list.	10	L2	3	2
6.	a) Define doubly linked list. Write C functions to insert a node at the following locations (i) at the front (ii) at a specific position (iii) at the end	10	L2	3	2
	b) Write a C program to insert a node at the beginning and to delete the node at the end of Singly Linked list.	10	L2	3	2
		UNIT-IV			
7.	a) Define threaded binary tree. Explain the categories	10	L2	4	2
	b) Define Binary Search Tree. Write a function to insert an item into an ordered binary search tree.	10	L2	4	2
8.	a) Discuss how the algorithm efficiency can be analysed.	10	L2	4	2
	b) Distinguish between (i) Algorithm and pseudo code (ii) Big oh and omega notations	10	L2	4	2
		UNIT-V			
9.	a) Explain Bubble Sort Algorithm with an example.	10	L2	5	1
	b) Explain Mergesort Algorithm with an example.	10	L2	5	2
10.	a) Explain Sequential Searching.	10	L2	5	1
	b) Explain Binary Search.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
